



Artwork Code : 2451000170-00				Artwork Req. No: 500000001686			
Quality of Paper/Board : Super Sun Shine				Size of Artwork (In mm) : 210x266			
Quality of Gum :				GSM of Paper/Board : 60 GSM			
Colour Code :	BLACK	Pantone	Pantone	Pantone	Pantone	Pantone	Pantone
Barcode Information:							
Barcode Scan Report:							
Packing: 100ml N. PVC bag plumat				Plant Location : MF 02			
Country: Malaysia				Language : English			
Ref. Code Creation/Blockage Note:				Artwork Control Key No.:			
“Controlled Copy” Holder		(1)	(2)	(3)	(4)	(5)	

Hepatobiliary disorders	Increased levels of hepatic transaminases		
Skin and subcutaneous tissue disorders	-	serious skin reactions (3)	Pruritus (2), Erythema (2)
General disorders and administration site conditions	Malaise	-	-

(1) Very rare cases of hypersensitivity reactions ranging from simple skin rash or urticaria to anaphylactic shock have been reported and require discontinuation of treatment.
(2) Isolated cases
(3) Very rare cases of serious skin reactions have been reported.
Cutaneous hypersensitivity reactions including skin rashes, angioedema, Stevens Johnson Syndrome/ Toxic Epidermal Necrolysis have been reported.
Frequent adverse reactions at injection site have been reported during clinical trials (pain and burning sensation).

Overdose

Symptoms

There is a risk of liver injury (including fulminant hepatitis, hepatic failure, cholestatic hepatitis, cytolytic hepatitis), particularly in elderly subjects, in young children, in patients with liver disease, in cases of chronic alcoholism, in patients with chronic malnutrition and in patients receiving enzyme inducers. Overdosing may be fatal in these cases.
Symptoms generally appear within the first 24 hours and comprise: nausea, vomiting, anorexia, pallor and abdominal pain. Immediate emergency measures are necessary in case of paracetamol overdose, even when no symptoms are present. Independent of the presence and severity of possible hepatic impairment, symptoms of acute renal impairment may develop in case of an overdose.

Overdose, 7.5 g or more of paracetamol in a single administration in adults or 140 mg/kg of body weight in a single administration in children, causes hepatic cytolysis likely to induce complete and irreversible necrosis, resulting in hepatocellular insufficiency, metabolic acidosis and encephalopathy which may lead to coma and death. Simultaneously, increased levels of hepatic transaminases (AST, ALT), lactate dehydrogenase and bilirubin are observed together with decreased prothrombin levels that may appear 12 to 48 hours after administration. Clinical symptoms of liver damage are usually evident initially after two days, and reach a maximum after 4 to 6 days.

Treatment

- Immediate hospitalisation.
- Before beginning treatment, take a blood sample for plasma paracetamol assay, as soon as possible after the overdose.
- The treatment includes administration of the antidote, N-acetylcysteine (NAC) by the intravenous or oral route, if possible before the 10th hour. NAC can, however, give some degree of protection even after 10 hours, but in these cases prolonged treatment is given.
- Symptomatic treatment.
- Hepatic tests must be carried out at the beginning of treatment and repeated every 24 hours. In most cases hepatic transaminases restitution to normal in one to two weeks with full return of normal liver function. In very severe cases, however, liver transplantation may be necessary.

Instructions for Use

Leniol should be administered in intravenous infusion over 15 minutes. To be used immediately after seal is opened and any leftover has to be discarded. Do not reuse any medications left over from previous dose. If you have the impression that Leniol 10mg/ml, solution for infusion is too strong or too weak, talk to your doctor immediately. No further dilution is required for this product as it is in intravenous infusion solution form and intended for single use only.

How supplied: Leniol is supplied in 100 mL Non PVC Bag in a moncarton with an insert.

Storage: Store at a temperature not exceeding 30°C, protected from light. Do not refrigerate.
To be used with non-pyrogenic I.V. administration set with aseptic technique.
Please use immediately after open. Discarded any unused portion.
For I.V. use only

Date of Revision: 06/07/2022

MAL No.: MAL22126057AZ



Manufactured by :
Otsuka Pharmaceutical India Private Limited
Survey No. 199 to 201 & 208 to 210,
Village-Vasana-Chacharwadi, Tal - Sanand,
Dist: Ahmedabad - 382 213, India.

Marketing Authorization Holder,
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2451000170

LENIOL - Paracetamol 1% w/v
Solution for Infusion

- Read the leaflet carefully before using
- Ask your pharmacist/ doctor for more information
- If you notice any side effects, please contact your doctor/pharmacist
- Keep away from Children

Composition

Each 100 ml Contains:

Paracetamol :1.0 g
Water for Injections :q.s.

Excipients: Sodium Metabisulfite, L-Histidine, Sodium Chloride, Sodium Hydroxide, Hydrochloric Acid, Water for Injections.
Description: Leniol is available as 100 mL Non PVC Bag.
It is a clear and colourless solution practically free from visible particles.

Indication:

LENIOL -Paracetamol 1% w/v Solution for Infusion is indicated for:

- short-term treatment of moderate pain, especially following surgery,
 - short-term treatment of fever,
- when administration by intravenous route is clinically justified by an urgent need to treat pain or hyperthermia and/or when other routes of administration are not possible.

Dose

The 100 ml pack is restricted to adults, adolescents and children weighing more than 33kg.

Posology:

Dosing based on patient weight (please see the dosing table here below)

Patient weight	Dose per administration	Volume per administration	Maximum volume of Paracetamol per administration based on upper weight limits of group (ml)***	Maximum daily dose**
>33 kg to ≤ 50 kg	15 mg/kg	1.5 ml/kg	75 ml	60 mg/kg not exceeding 3 g
>50 kg with additional risk factors for hepatotoxicity	1 g	100 ml	100 ml	3 g
>50 kg and no additional risk factors for hepatotoxicity	1 g	100 ml	100 ml	4 g

** Maximum daily dose:

The maximum daily dose as presented in the table above is for patients that are not receiving other paracetamol containing products and should be adjusted accordingly taking such products into account.

*** Patients weighing less will require smaller volumes

The minimum interval between each administration must be at least 4 hours.

The minimum interval between each administration in patients with severe renal insufficiency must be at least 6 hours.

No more than 4 doses to be given in 24 hours.

Severe renal insufficiency:

It is recommended, when giving paracetamol to patients with severe renal impairment (creatinine clearance ≤ 30 ml/min), to reduce the dose and increase the minimum interval between each administration to 6 hours.

Adults with hepatocellular insufficiency, chronic alcoholism, chronic malnutrition (low reserves of hepatic glutathione), dehydration:

The maximum daily dose must not exceed 3000 mg

"This product is only available for 100ml. Please look for alternative for the paediatric dosage"

Method of administration

Take care when prescribing and administering LENIOL -Paracetamol 1% w/v Solution for Infusion to avoid dosing errors due to confusion between milligram (mg) and millilitre (ml), which could result in accidental overdose and death. Take care to ensure the proper dose is communicated and dispensed. When writing prescriptions, include both the total dose in mg and the total dose in volume. Take care to ensure the dose is measured and administered accurately.

Intravenous use.

The paracetamol solution is administered as a 15-minute intravenous infusion.



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For single use only. Any unused solution should be discarded. Before administration, the product should be visually inspected for any particulate matter and discolouration. Only to be used if solution is clear, colourless to slightly pinkish-orangish (perception may vary) and the container and its closure are undamaged. Colouration can intensify over time without adversely affecting the quality of the product. As for all solutions for infusion presented in containers with air space inside, it should be remembered that close monitoring is needed notably at the end of the infusion, regardless of administration route. This monitoring at the end of the infusion applies particularly for central route infusions, in order to avoid air embolism.

Contraindication

- Hypersensitivity to paracetamol, propacetamol hydrochloride (prodrug of paracetamol) or to any of the excipients mentioned above.
- Cases of severe hepatocellular insufficiency.

Pregnancy and lactation

Pregnancy:
Clinical experience of the intravenous administration of paracetamol is limited. However, epidemiological data from the use of oral therapeutic doses of paracetamol indicate no undesirable effects in pregnancy or on the health of the foetus / newborn infant. Prospective data on pregnancies exposed to overdoses did not show any increase in the risk of malformation. No reproductive studies with the intravenous form of paracetamol have been performed in animals. However, studies with the oral route did not show any malformation or foetotoxic effects. Nevertheless, LENIOL -Paracetamol 1% w/v Solution for Infusion should only be used during pregnancy after a careful benefit-risk assessment. In this case, the recommended posology and duration must be strictly observed.

Lactation:
After oral administration, paracetamol is excreted into breast milk in small quantities. No undesirable effects on nursing infants have been reported. Consequently, LENIOL -Paracetamol 1% w/v Solution for Infusion may be used in breastfeeding women.

Drug Interactions

- Probenecid causes an almost 2-fold reduction in clearance of paracetamol by inhibiting its conjugation with glucuronic acid. A reduction of the paracetamol dose should be considered for concomitant treatment with probenecid,
- Salicylamide may prolong the elimination t1/2 of paracetamol,
- Caution should be paid to the concomitant intake of enzyme-inducing substances.
- Concomitant use of paracetamol (4 g per day for at least 4 days) with oral anticoagulants may lead to slight variations of INR values. In this case, increased monitoring of INR values should be conducted during the period of concomitant use as well as for 1 week after paracetamol treatment has been discontinued.

Pharmacodynamic effects

LENIOL -Paracetamol 1% w/v Solution for Infusion provides onset of pain relief within 5 to 10 minutes after the start of administration. The peak analgesic effect is obtained in 1 hour and the duration of this effect is usually 4 to 6 hours. LENIOL -Paracetamol 1% w/v Solution for Infusion reduces fever within 30 minutes after the start of administration with a duration of the antipyretic effect of at least 6 hours.

Pharmacokinetics

Adults

Absorption:

Paracetamol pharmacokinetics is linear up to 2 g after single administration and after repeated administration during 24 hours. The bioavailability of paracetamol following infusion of 500 mg and 1 g of paracetamol is similar to that observed following infusion of 1 g and 2 g propacetamol (corresponding to 500 mg and 1 g paracetamol respectively). The maximal plasma concentration (Cmax) of paracetamol observed at the end of 15-minutes intravenous infusion of 500 mg and 1 g of paracetamol is about 15 µg/mL and 30 µg/mL respectively.

Distribution:

The volume of distribution of paracetamol is approximately 1 L/kg. Paracetamol is not extensively bound to plasma proteins. Following infusion of 1 g paracetamol, significant concentrations of paracetamol (about 1.5 µg/mL) were observed in the Cerebro Spinal Fluid as and from the 20th minute following infusion.

Metabolism:

Paracetamol is metabolised mainly in the liver following two major hepatic pathways: glucuronic acid conjugation and sulphuric acid conjugation. The latter route is rapidly saturable at doses that exceed the therapeutic doses. A small fraction (less than 4%) is metabolised by cytochrome P450 to a reactive intermediate (N-acetyl benzoquinone imine) which, under normal conditions of use, is rapidly detoxified by reduced glutathione and eliminated in the urine after conjugation with cysteine and mercapturic acid. However, during massive overdosing, the quantity of this toxic metabolite is increased.

Elimination:

The metabolites of paracetamol are mainly excreted in the urine. 90% of the dose administered is excreted in 24 hours, mainly as glucuronide (60-80%) and sulphate (20-30%) conjugates. Less than 5% is eliminated unchanged. Plasma half-life is 2.7 hours and total body clearance is 18 L/h.

Neonates, infants and children

The pharmacokinetic parameters of paracetamol observed in infants and children are similar to those observed in adults, except for the plasma half-life that is slightly shorter (1.5 to 2 h) than in adults. In neonates, the plasma half-life is longer than in infants i.e. around 3.5 hours. Neonates, infants and children up to 10 years excrete significantly less glucuronide and more sulphate

conjugates than adults.
*Table. Age related pharmacokinetic values (standardized clearance, *CLstd/Foral × (1.h-1 × 70 kg-1), are presented below.*

Age	Weight (kg)	CLstd/Foral (1.h-1 × 70 kg-1)
40 weeks post-conception	3.3	5.9
3 months postnatal	6	8.8
6 months postnatal	7.5	11.1
1 year postnatal	10	13.6
2 years postnatal	12	15.6
5 years postnatal	20	16.3
8 years postnatal	25	16.3

*Clstd is the population estimate for CL

Special populations

Renal impairment

In cases of severe renal impairment (creatinine clearance 10-30 mL/min), the elimination of paracetamol is slightly delayed, the elimination half-life ranging from 2 to 5.3 hours. For the glucuronide and sulphate conjugates, the elimination rate is 3 times slower in subjects with severe renal impairment than in healthy subjects. Therefore, it is recommended, when giving paracetamol to patients with severe renal impairment (creatinine clearance ≤ 30 mL/min), to increase the minimum interval between each administration to 6 hours.

Elderly subjects

The pharmacokinetics and the metabolism of paracetamol are not modified in elderly subjects. No dose adjustment is required in this population.

Warnings and Precautions

WARNING

This preparation contains PARACETAMOL.
Do not take any other paracetamol containing medicines at the same time.

Allergy alert: Paracetamol may cause severe skin reactions. Symptoms may include skin reddening, blisters or rash. These could be signs of a serious condition. If these reactions occur, stop use and seek medical assistance right away. This preparation contains Sodium Metabisulphite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulphites.

RISK OF MEDICATION ERRORS

Take care to avoid dosing errors due to confusion between milligram (mg) and milliliter (ml), which could result in accidental overdose and death

Prolonged or frequent use is discouraged. It is recommended that a suitable analgesic oral treatment will be used as soon as this route of administration is possible. Doses higher than those recommended entail the risk of very serious liver damage. Clinical signs and symptoms of liver damage (including fulminant hepatitis, hepatic failure, cholestatic hepatitis, cytolytic hepatitis) are usually first seen after two days of drug administration with a peak seen, usually after 4–6 days. Treatment with antidote should be given as soon as possible. Paracetamol should be used with caution in cases of:

- hepatocellular insufficiency
- severe renal insufficiency (creatinine clearance ≤ 30 ml/min)
- chronic alcoholism
- chronic malnutrition (low reserves of hepatic glutathione)
- dehydration
- patients suffering from a genetically caused G-6-PD deficiency (favism), the occurrence of a haemolytic anaemia is possible due to the reduced allocation of glutathione following the administration of paracetamol.

Side Effects

As with all paracetamol products, adverse drug reactions are rare (≥ 1/10 000 to <1/1 000) or very rare (<1/10 000). They are described below:

System Organ Class	Rare (≥1/10 000 to <1/1 000)	Very rare (<1/10 000)	Not known (cannot be estimated from the available data)
Blood and lymphatic system disorders	-	Thrombocytopenia, Leucopenia, Neutropenia	-
Immune system disorders	-	Hypersensitivity reaction (1, 3)	-
Cardiac disorders	-	-	Tachycardia (2)
Vascular disorders	Hypotension	-	Flushing (2)